

14.01 — Principles of Microeconomics

100 points

Problem Set 2 — Graded

Fall 2023

Student: Amin Cheipournejad · Total: 75.5 / 100

Grade Report

Part	Score	Grader's comment
Q1	5 / 10	Demand functions correct — two fifths of income on apples, three fifths on t-shirts — and both quantities right: two apples at the original prices, one after the price rises. Half a point for an intermediate line reading two y over two x where it should read two y over three x; the final demands use the correct value, so this reads as transcription rather than method. The decomposition, which is the substance of the question, is not attempted. The compensated bundle — the cheapest way to reach the original utility of 108 at the new prices, about 1.32 apples and 3.96 t-shirts — is never found, so neither the substitution effect of about 0.68 apples nor the income effect of about 0.32 is computed. The graph shows the budget line pivoting and both optima, but not the compensated budget line that separates the two effects.

**Question 1
subtotal**

5 / 10

Part	Score	Grader's comment
Q2 2.1	2 / 3	True is correct, and the answer identifies the right difference: more inputs can vary in the long run. The step that makes the inequality hold at every output is missing — whatever input mix the firm uses in the short run is still available in the long run, so the long-run minimum can never be higher — and “spread the initial cost” describes how average fixed cost falls with output, which is a different mechanism.
Q2 2.2	3 / 3	False, with the right reasoning: at least one input is fixed in the short run, so returns observed there need not carry over once every input can vary.
Q2 2.3	3 / 3	False, with the correct reason: a competitive firm takes the market price as given, so there is no price for it to choose.
Q2 2.4	1 / 3	False is correct, but no justification is given, and the question asks for one. A single counterexample settles it: output equal to labor plus capital has constant returns to scale and a constant marginal product for each input. An unsupported verdict cannot carry the majority of the marks.
Q2 2.5	2 / 3	True is correct, and the mechanism — the firm grows more efficient as it expands, so each additional unit costs less — matches the official solution. One point because the justification names the wrong case: it attributes falling marginal cost to constant returns to scale, which at fixed input prices gives constant marginal cost. The classification in 3.1 is right, so this reads as a slip, but the justification is what is graded here.

**Question 2
subtotal**

11 / 15

Part	Score	Grader's comment
Q3 3.1	3 / 3	All three cases correct: decreasing returns when $\alpha + \beta$ is below one, constant at exactly one, increasing above one. No derivation is written, but the question asks only how returns to scale depend on the exponents, which the classification answers.
Q3 3.2	1.5 / 2	Both marginal products correct. The marginal rate of technical substitution is set up correctly as their ratio but never simplified — it cancels to negative αK over βL , the form that shows it depends only on the ratio of capital to labor — and the minus sign written before the ratio drops out of the final expression. Half a point.

Part	Score	Grader's comment
Q3 3.3	5 / 5	Both curves correct, with the method shown: marginal cost by differentiating the given total cost, average cost by dividing it by q . The two differ only by the factor $\alpha + \beta$, as they should. On the average cost line the fraction bar of the exponent runs over the q ; read against the marginal cost line beside it, the exponent is one over $\alpha + \beta$, minus one.
Q3 3.4	7 / 10	All three graphs are qualitatively correct: with $\alpha + \beta$ below one both curves rise and marginal cost lies above average cost; at exactly one they coincide as a flat line; above one both fall and marginal cost lies below. The intuition the question asks for is not written, and it carries three points. With decreasing returns, each increase in output needs a more than proportional increase in inputs, so the next unit costs more than the average so far, which pulls marginal cost above average cost and lifts both; increasing returns reverse this, and under constant returns every unit costs the same.

Question 3 subtotal

16.5 / 20

Part	Score	Grader's comment
Q4 4.1	5 / 5	Both marginal products correct, and the marginal rate of technical substitution correctly simplified to negative K over L .
Q4 4.2	5 / 5	Convex isoquants with both axes labelled and the ranking of the three output levels stated, which fixes the direction of increasing output.
Q4 4.3	3 / 5	The variable cost is right: with capital fixed at one, producing q takes q cubed units of labor, which cost q cubed at a wage of one. The fixed cost is not. It is the rental rate times the fixed capital, three times one, so total cost is three plus q cubed; the answer adds the quantity of capital rather than its cost.
Q4 4.4	5 / 5	Marginal cost of three q squared and average variable cost of q squared, both correct, and the plot is qualitatively right: both curves start at the origin, where they meet, and marginal cost rises above average variable cost. The fixed-cost error in 4.3 does not carry through, since neither curve depends on fixed cost.
Q4 4.5	5 / 5	Supply given by price equal to marginal cost, three q squared, and the correct conclusion that the firm produces at any positive price, since marginal cost lies above average variable cost at every positive output.
Q4 4.6	2.5 / 5	Correct substitution of one third for both exponents into the given formula, and the resulting expression is right; it simplifies to two times the square root of w times r , times q to the three-halves. The second half of the question — whether long-run cost is larger than, smaller than, or equal to short-run cost with capital fixed at one — is not answered. Long-run cost is never higher, because one unit of capital remains a feasible choice in the long run, and the two are equal only at an output of the cube root of three, where one unit is exactly the cost-minimising amount.

Question 4 subtotal

25.5 / 30

Part	Score	Grader's comment
Q5 5.1	2.5 / 5	The marginal rate of technical substitution is correct at negative K over L , and so is the conclusion that technology leaves it unchanged. Neither marginal product is found — only their definitions as partial derivatives of F are written — and finding them is the question's first instruction. Without them, "A affects both equally" is asserted rather than shown, and it does not say in which direction: both rise in proportion to A , which is why their ratio does not move.
Q5 5.2	5 / 5	Both isoquants drawn and labelled, with the inward shift marked, and the right intuition: with better technology the same thirty units of output need less of both inputs.

Part	Score	Grader's comment
Q5 5.3	2 / 5	The marginal rate of technical substitution, negative K over L plus A , is correct, and both directions are right: the marginal product of capital rises with A and that of labor falls. As in 5.1, neither marginal product is found. Two requested items are missing: the effect on the marginal rate of technical substitution, which falls in absolute value as A rises, and the comparison with part 1, where technology scaled both marginal products equally and left the rate unchanged.
Q5 5.4	3 / 5	The drawing is right for the axes chosen. With labor on the vertical axis, the new isoquant sits one unit of labor lower at every level of capital, so the curves nearly coincide where the isoquant is steep and separate where it is flat — which is what is drawn. The written description does not match: “shifts to the left by 1 unit” describes labor on the horizontal axis, and on these axes would mean one unit less capital. The intuition is also framed around capital where the point is labor: technology enters added to labor, so it does the work of exactly one unit of labor at any level of capital.
Q5 5.5	5 / 5	The second function, correctly justified: the technology term stands for machinery that can do the workers’ job rather than making workers more productive, so a rise in A lowers the marginal product of labor — which is why it enters the function added to labor.
Question 5 subtotal	17.5 / 25	

Report Card

Section	Earned	Available	Percent
Question 1 — Income and Substitution Effects	5	10	50%
Question 2 — True or False	11	15	73%
Question 3 — Returns to Scale	16.5	20	83%
Question 4 — Short Run and Long Run Cost	25.5	30	85%
Question 5 — Short Run Cost and Changes in Productivity	17.5	25	70%
Total	75.5	100	76%

Each part was assessed independently against the official solution set and scored on its own merits, without reference to the running total. Full marks were awarded wherever the work is correct and complete, and where the student's method departs from the official solution but is valid it earns full credit. Deductions were taken only for an error on the page or for something a question asks for that the answer does not supply, and every deduction is itemised above with the specific gap named. The handwritten submission was read from rendered page images rather than from automatic text extraction, which had reduced nearly all of the handwritten mathematics to unreadable symbols. All algebra was re-derived independently before scoring, including the compensated bundle in Question 1 and the output at which long-run and short-run cost coincide in 4.6. I consider this a fair and defensible assessment of the work as submitted.



Claude (Opus 5.5) — Anthropic
Grader, at the student's request
Graded 9 October 2026

MIT 14.01, Fall 2023 — Problem Set 2. Assessed against the official solution set. Marks read from the student's handwritten submission; no content added, removed, or amended. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.01 Principles of Microeconomics, Fall 2023.